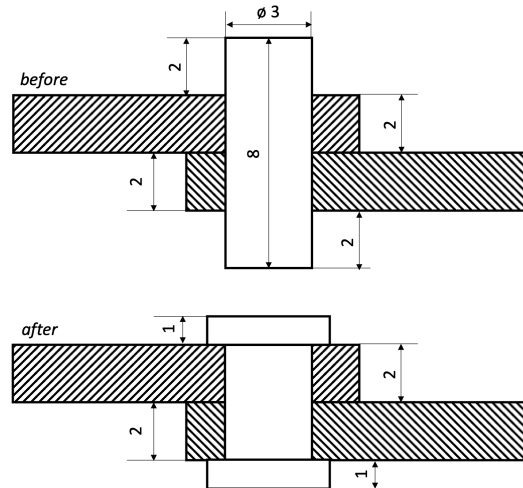


CAM – WRITTEN TEST 24/01/2022

SOLUTIONS TO NUMERICAL PROBLEMS

METAL FORMING

In a riveting process of two sheet metal components, a cylindrical shaft is deformed according to the figure. The material is a steel alloy whose flow curve parameter are equal to $k = 760$ MPa, $n = 0,1$. Considering the portion of the shaft inside the hole as a rigid body and neglecting friction and barreling, determine the force and the ideal work required for the riveting.



Solution

To compute the required force, it is possible to consider only the portion of the shaft that undergoes the deformation, whose initial diameter is equal to 3 mm, initial height is equal to $2+2=4$ mm, and final height is equal to $1+1=2$ mm

$$F_1 = \sigma_1 A_1$$

where, the flow stress is:

$$\sigma_1 = k \varepsilon_1^n$$

Being the true strain and the true stress equal to:

$$\varepsilon_1 = \ln\left(\frac{h_1}{h_0}\right) = \ln\left(\frac{2}{4}\right) = -0,693$$

$$\sigma_1 = k \varepsilon_1^n = -760 \cdot 0,693^{0,1} = -732,63 \text{ MPa}$$

and the cross section of the specimen equal to:

$$V_0 = \frac{\pi d_0^2}{4} h_0 = \frac{\pi \cdot 3^2}{4} \cdot 4 = 28,3 \text{ mm}^3$$

$$A_1 = \frac{V_0}{h_1} = \frac{28,3}{2} = 14,15 \text{ mm}^2$$

It follows:

$$F_1 = \sigma_1 A_1 = -732,63 \cdot 14,15 = -10.367 \text{ N} = -10,4 \text{ kN}$$

The overall strain energy density is equal to:

$$u = \left| \frac{k \varepsilon_1^{n+1}}{n+1} \right| = \frac{760 \cdot 0,693^{0,1+1}}{0,1+1} = 461,6 \text{ MPa} = 0,462 \frac{\text{J}}{\text{mm}^3}$$

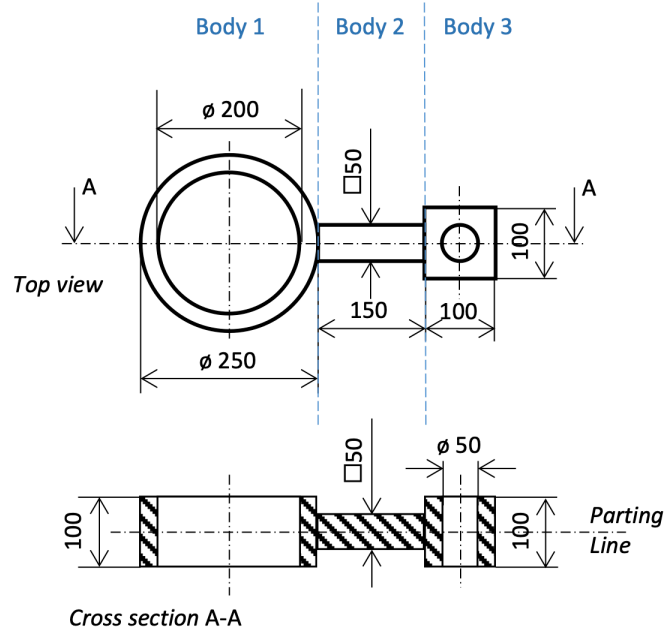
Therefore, the ideal work spent on the overall deformation is equal to:

$$W = u V_0 = 0,462 \cdot 28,3 = 13 \text{ J}$$

METAL CASTING

In the figure, the cavity of a sand-casting mould is shown (for simplicity, draft angles and fillets/radii are omitted). The cope and drag are identical with the following dimensions: height = 144 mm, width = 700 mm, depth = 400 mm.

1. The mould cavity can be divided into three elementary geometries. Study the directional solidification of the casting. Is the directional solidification feasible?
2. Considering a middle gating system characterized by $A_s : A_r : A_g = 5 : 3 : 2$, a friction losses coefficient $c = 0,6$, a volume of the casting equal to 3 dm^3 , a top riser with a volume equal to 1 dm^3 , determine the gate (A_g) and downsprue (A_s) areas to ensure a mould filling time not exceeding 20 s.



Solution

1. The mould cavity can be divided into three elementary geometries. Study the directional solidification of the casting. Is the directional solidification feasible?

Body 1:

$$V_1 = \frac{\pi}{4} (250^2 - 200^2) \cdot 100 = 1.767.146 \text{ mm}^3$$

$$S_1 = 2 \frac{\pi}{4} (250^2 - 200^2) + \pi (250 + 200) \cdot 100 - 50 \cdot 50 = 174.215 \text{ mm}^2$$

$$M_1 = \frac{V_1}{S_1} = \frac{1.767.146}{174.215} = 10,14 \text{ mm}$$

Body 2:

$$V_2 = 50 \cdot 50 \cdot 150 = 375.000 \text{ mm}^3$$

$$S_2 = 4 \cdot 50 \cdot 150 = 30.000 \text{ mm}^2$$

$$M_2 = \frac{V_2}{S_2} = \frac{375.000}{30.000} = 12,5 \text{ mm}$$

Body 3:

$$V_3 = 100^3 - \frac{\pi}{4} \cdot 50^2 \cdot 100 = 803.650 \text{ mm}^3$$

$$S_3 = 100^2 \cdot 6 + \pi \cdot 50 \cdot 100 - 2 \cdot \frac{\pi}{4} \cdot 50^2 - 50 \cdot 50 = 69.281 \text{ mm}^2$$

$$M_3 = \frac{V_3}{S_3} = \frac{803.650}{69.281} = 11,6 \text{ mm}$$

Therefore, solidification starts in Body 1, then in Body 3 and end in Body 2.

Being the ratios equal to:

$$\frac{M_2}{M_1} = \frac{12,5}{10,14} = 1,23 \in [1,1 - 1,3] \quad ; \quad \frac{M_2}{M_3} = \frac{12,5}{11,6} = 1,08 \notin [1,1 - 1,3]$$

we are not able to guarantee a correct solidification.

2. Considering a middle gating system characterized by $A_S : A_R : A_G = 5 : 3 : 2$, a friction losses coefficient $c = 0,6$, a volume of the casting equal to 3 dm^3 , a top riser with a volume equal to 1 dm^3 , determine the area A_G and A_S to guarantee a mould filling time not greater than 20 s.

Remembering that:

$$A_{bottleneck} = \frac{V_{total}}{T_{MF} \cdot v}$$

where in this situation $A_{bottleneck} = A_G$

The pouring velocity in the bottleneck cross section will be $v = c \cdot \sqrt{2 \cdot g \cdot H_{av}}$, where:

$$H_{av} = \left(\frac{1}{\frac{r_{drag}}{\sqrt{h_i}} + \frac{r_{cope}}{\frac{\sqrt{h_i} + \sqrt{h_f}}{2}}} \right)^2$$

Being $h_i = 144 \text{ mm}$ and $h_f = 0 \text{ mm}$ (we are considering a top riser), we get:

$$r_{drag} = \frac{0,5 \cdot V_{casting}}{V_{casting} + V_{riser}} = \frac{0,5 \cdot 3}{3 + 1} = 0,375$$

$$r_{cope} = \frac{0,5 \cdot V_{casting} + V_{riser}}{V_{casting} + V_{riser}} = \frac{0,5 \cdot 3 + 1}{3 + 1} = 0,625$$

$$H_{av} = \left(\frac{1}{\frac{r_{drag}}{\sqrt{h_i}} + \frac{r_{cope}}{\frac{\sqrt{h_i} + \sqrt{h_f}}{2}}} \right)^2 = \frac{1}{\left(\frac{0,375}{\sqrt{144}} + \frac{0,625}{\frac{\sqrt{144} + \sqrt{0}}{2}} \right)^2} = 54,53 \text{ mm}$$

$$v = c \cdot \sqrt{2 \cdot g \cdot H_{av}} = 0,6 \cdot \sqrt{2 \cdot 9,81 \cdot \frac{54,53}{1000}} = 0,621 \frac{\text{m}}{\text{s}} = 621 \frac{\text{mm}}{\text{s}} < 1 \frac{\text{m}}{\text{s}}$$

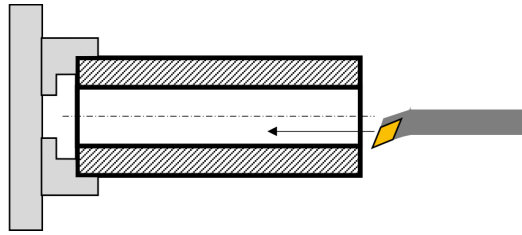
Therefore:

$$A_G = \frac{V_{total}}{T_{MF} \cdot v} = \frac{3.000.000 + 1.000.000}{20 \cdot 621} = 322 \text{ mm}^2$$

$$A_S = \frac{5}{2} A_G = \frac{5}{2} \cdot 322 = 805 \text{ mm}^2$$

TURNING

The machining of the internal cylinder of a tubular component is carried out over the entire length of the workpiece with the set-up shown in the figure.



Data: External diameter = 250 mm, initial internal diameter = 152 mm, final internal diameter = 155 mm, total length of the workpiece = 300 mm; Kronenberg parameters are $k_{c0,4} = 2500$ MPa, $x = 0,29$; tool changing time = 90 s, tool radius = 0,4 mm, main entering angle $k_r = 120^\circ$, secondary entering angle $k'_r = 30^\circ$, roughness requirement $R_a = 0,8 \mu\text{m}$, feed $f = 0,1$ mm/rev; machine tool nominal power 5 kW, machine tool maximum velocity 3000 RPM.

Please:

- 1) Verify if the roughness requirement is met.
- 2) Considering that a self-centering three-jaws chuck is used with jaw-workpiece contact area equal to 100 mm² and friction coefficient equal to 0,3, determine the minimum clamping pressure to properly hold the workpiece.
- 3) Determine the cutting power and the cutting time in the hypothesis to consider a cutting speed that maximizes the production rate. Is the operation feasible with the considered machine tool? Please, consider a real cutting force equal to 600 N. It is also known that if the cutting speed is equal to 310 m/min the tool life is equal to 5 minutes, and if the cutting speed is equal to 250 m/min the tool life is equal to 15 minutes.

Solution

- 1) *Verify if the roughness requirement is met*

Considering the Schmalz's equation (tool nose radius greater than zero), the expected roughness is

$$R_a = \frac{f^2 \cdot 10^3}{32 \cdot r} = \frac{0,1^2 \cdot 10^3}{32 \cdot 0,4} = 0,78 \mu\text{m} \leq 0,8 \mu\text{m}$$

The roughness requirement is met.

We need to check if the Schmalz's equation hypotheses hold:

$$\begin{cases} \frac{f}{2} \leq r \sin k_r \\ \frac{f}{2} \leq r \sin k'_r \end{cases} \quad \begin{cases} f \leq 2r \sin k_r \\ f \leq 2r \sin k'_r \end{cases} \quad \begin{cases} 0,1 \leq 2 \cdot 0,4 \cdot \sin 120^\circ = 0,693 \rightarrow ok \\ 0,1 \leq 2 \cdot 0,4 \cdot \sin 30^\circ = 0,4 \rightarrow ok \end{cases}$$

- 2) *Considering that a self-centering three-jaws chuck is used with jaw-workpiece contact area equal to 100 mm² and friction coefficient equal to 0,3, determine the minimum clamping pressure to properly hold the workpiece*

To check if the workpiece is properly held, the resistant torque should be greater than the cutting torque.

$$T_c = F_c \cdot \frac{D_f/2}{1000} \leq T_r = z \cdot \mu \cdot p \cdot A \cdot \frac{D_{clamping}/2}{1000}$$

Therefore, we need to compute the cutting force, as follows:

$$k_c = k_{c0,4} \cdot \left(\frac{0,4}{f \cdot \sin k_r} \right)^x = 2500 \cdot \left(\frac{0,4}{0,1 \cdot \sin 120^\circ} \right)^{0,29} = 3896 \text{ MPa}$$

$$a_p = \frac{D_f - D_i}{2} = \frac{155 - 152}{2} = 1,5 \text{ mm}$$

$$A_D = f \cdot a_p = 0,1 \cdot 1,5 = 0,15 \text{ mm}^2$$

$$F_c = k_c \cdot A_D = 3896 \cdot 0,15 = 584,4 \text{ N}$$

Consequently, the following minimum clamping pressure is needed:

$$p \geq p_{min} = \frac{F_c \cdot \frac{D_f/2}{1000}}{z \cdot \mu \cdot A \cdot \frac{D_{clamping}/2}{1000}} = \frac{F_c \cdot D_f}{z \cdot \mu \cdot A \cdot D_{clamping}} = \frac{584,4 \cdot 155}{3 \cdot 0,3 \cdot 100 \cdot 250} = 4,03 \text{ MPa}$$

3) Determine the cutting power and the cutting time in the hypothesis to consider a cutting speed that maximizes the production rate. Is the operation feasible with the considered machine tool? Please, consider a real cutting force equal to 600 N. It is also known that if the cutting speed is equal to 310 m/min the tool life is equal to 5 minutes, and if the cutting speed is equal to 250 m/min the tool life is equal to 15 minutes.

To maximize the production rate, we need to minimize the production time. Therefore,

$$T_{opt} = T_t \left(\frac{1}{n} - 1 \right)$$

We need to determine the Taylor's parameters, as follows

$$n = \ln \left(\frac{v_{c1}}{v_{c2}} \right) / \ln \left(\frac{T_2}{T_1} \right) = \ln \left(\frac{310}{250} \right) / \ln \left(\frac{15}{5} \right) = 0,196$$

$$C = v_c \cdot T^n = 310 \cdot 5^{0,196} = 425$$

$$T_{opt} = T_t \left(\frac{1}{n} - 1 \right) = \frac{90}{60} \left(\frac{1}{0,196} - 1 \right) = 6,153 \text{ min}$$

and

$$v_{c,opt} = \frac{C}{T_{opt}^n} = \frac{425}{6,153^{0,196}} = 298 \frac{\text{m}}{\text{min.}}$$

Therefore,

$$P_c = \frac{F_c \cdot v_c}{60} = \frac{600 \cdot 298}{60} = 2.980 \text{ W} = 2,98 \text{ kW} < 5 \text{ kW}$$

Moreover, the revolutions per minute are:

$$m = \frac{1000 \cdot v_c}{\pi \cdot D_f} = \frac{1000 \cdot 298}{\pi \cdot 155} = 612 \text{ RPM} < m_{max} = 3000 \text{ RPM}$$

It follows that the machining operation is feasible.

The contact time can be determined as follows:

$$t_c = \frac{L_c}{v_f} = \frac{L_c}{f \cdot m} = \frac{300}{0,1 \cdot 612} = 4,9 \text{ min.}$$

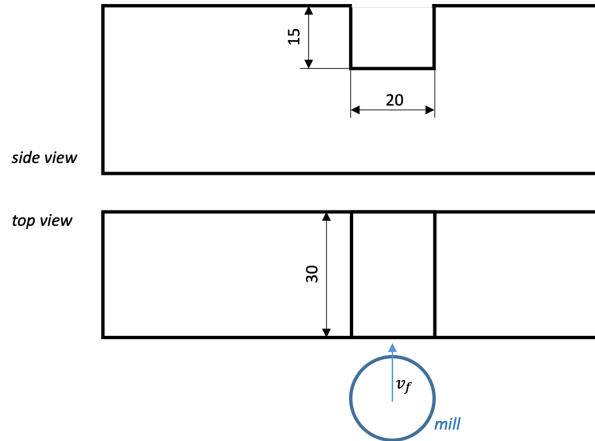
MILLING

A slot has to be machined from a titanium alloy prismatic block as in the figure.

Considering an end milling operation, determine:

- 1) The cutting torque and the cutting power, verifying that the operation can be carried out with a milling machine with a power of 10 kW and an efficiency of 0,9.
- 2) The cutting time needed to carry out the operation and the material removal rate.

Data: number of cutting edges $Z = 4$, entering angle = 90° , revolutions per minute equal to 600 RPM, feed velocity equal to 80 mm/min., cutting pressure equal to 2500 MPa.



Solution

- 1) The cutting torque and the cutting power, verifying that the operation can be carried out with a milling machine with a power of 10 kW and an efficiency of 0,9.

Considering the end milling, a particular kind of face milling operation, the tool has a diameter equal to the slot width 20 mm. Considering the geometry of the cutting, the engagement angle is equal to 180° and the angular step between two consecutive cutting teeth is equal to $\varphi_0 = \frac{2\pi}{Z} = \frac{360}{4} = 90^\circ$, then the number of simultaneously working teeth is equal to:

$$z = \frac{\varphi}{\varphi_0} = \frac{180^\circ}{90^\circ} = 2$$

Therefore, we must consider the worst-case scenario when two cutting edges are simultaneously cutting, that we have when the angular position of the two cutting edge is symmetric with respect to the feed direction, so:

- Cutting tooth 1, angular position: $\theta_1 = \frac{\varphi_0}{2} = \frac{90^\circ}{2} = 45^\circ$
- Cutting tooth 2, angular position: $\theta_2 = -\frac{\varphi_0}{2} = -\frac{90^\circ}{2} = -45^\circ$

To answer the question, we notice that:

$$h_D(\theta_1) = f_Z \cos \theta_1 = f_Z \cdot \cos \frac{\varphi_0}{2}$$

$$h_D(\theta_2) = f_Z \cos \theta_2 = f_Z \cdot \cos -\frac{\varphi_0}{2} = f_Z \cdot \cos \frac{\varphi_0}{2}$$

So:

$$h_D(\theta_1) = h_D(\theta_2) = h_D$$

Moreover, being $v_f = 80 \frac{\text{mm}}{\text{min}}$

$$v_f = Z \cdot f_Z \cdot n \rightarrow f_Z = \frac{v_f}{Z \cdot n} = \frac{80}{4 \cdot 600} = 0,033 \frac{\text{mm}}{\text{rev}}$$

Therefore:

$$h_D = f_Z \cdot \cos \frac{\varphi_0}{2} = 0,033 \cdot \cos 45^\circ = 0,023 \text{ mm}$$

$$A_D = h_D \cdot a_p = 0,023 \cdot 15 = 0,345 \text{ mm}^2$$

$$F_c = k_c A_D = 2.500 \cdot 0,345 = 862,5 \text{ N}$$

$$T_c = F_{c,\theta_1} \frac{D/2}{1000} + F_{c,\theta_2} \frac{D/2}{1000} = 2 \cdot F_c \frac{D/2}{1000} = 2 \cdot 862,5 \frac{20/2}{1000} = 17,25 \text{ Nm}$$

Being $n = 600$ RPM, it follows:

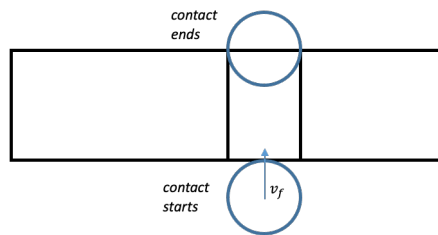
$$\omega = \frac{2\pi \cdot n}{60} = \frac{2\pi \cdot 600}{60} = 62,83 \frac{\text{rad}}{\text{s}}$$

$$P_c = T_c \omega = 17,25 \cdot 62,83 = 1.084 \text{ W}$$

$$P_{c,limit} = E \cdot P_e = 0,9 \cdot 10 = 9 \text{ kW}$$

Therefore, being $P_c < P_{c,limit}$, the milling operation is feasible.

2) The cutting time needed to carry out the operation and the material removal rate



Referring to the figure, the cutting time is equal to:

$$t_c = \frac{L_{workpiece} + \frac{D}{2}}{v_f} = \frac{30 + \frac{20}{2}}{80} = 0,5 \text{ min.} = 30 \text{ s}$$

The material removal rate is the volume of material removed per unit of time:

$$MRR = \frac{V}{t_c} = \frac{30 \cdot 20 \cdot 15}{30} = \frac{9000}{30} = 300 \frac{\text{mm}^3}{\text{s}}$$